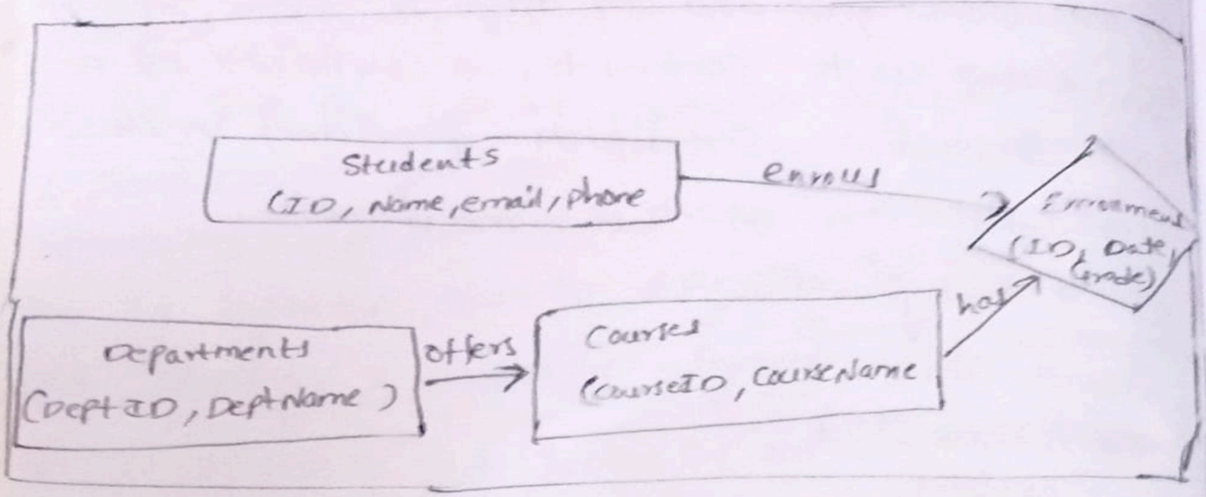
② ER - Diagram :



③ Exercise :

Consider a banking transaction : Transfer ₹1000 from Account A to Account B.
Explain how ACID properties (Atomicity, Consistency, Isolation, Durability) apply in this case.

Ans:- **Atomicity** :

(1) The transaction must either fully happen or not happen at all.

(2) Example : If ₹1000 is deducted from Account A but not added to Account B due to a failure, the transaction is rolled back.

② **Consistency** :-

• The database must remain in valid state before & after the transaction.

• Ex:- The total balance of Account A + Account B must remain the same before & after the transfer. No money is lost or magically created.

3. Isolation:-

• Multiple transactions should not interfere with each other.

• Ex:- if two people try to transfer money at the same time, each transaction is executed as if it's only one happening - no mix-up of balances.

4. Durability:-

• Once transaction is committed, the changes are permanent even if the system crashes.

Ex:- After the ₹ 1000 transfer is completed & committed, the updated balances of both accounts will be stored safely in the db.

4. Normalization

Exercise:

Given the following flat table, normalize it to Third normal form (3NF):

| StudentID | StudentName | DeptName | AdvisorName |

Explain each normalization step briefly 1NF → 2NF → 3NF

Sol:

1NF: (First Normal Form)

① Rule: Removing repeating groups → each cell should hold atomic (single) values.

In this case, the table is already atomic (no multi-valued attributes).

2NF:

Rule: Eliminate partial dependency → non key attributes should depend on the whole primary key

- primary key here: StudentID
  - Issues
    - DeptName depends on Student's department
    - AdvisorName depends on department, not the student

so we split

StudentsTable

|StudentID| StudentName| DeptName|

Departments Table

|DeptName| AdvisorName|

3NF :

Rule: Eliminate transitive dependency → Non-key attribute should not depend on other non-key attributes.

- In students table, DeptName → AdvisorName

Final Structure

Students Table

|Student ID| StudentName| DeptName|

Department Table

|DeptName| AdvisorID|

Advisors Table

|AdvisorID| AdvisorName